

Sergey Zimakov

Backend & Embedded Software Engineer

✉ zimakovs93@gmail.com | ☎ +972544366021

🌐 [LinkedIn](#) | 📄 [GitHub](#) | 🌐 [Web Site](#)

Profile

Experienced Backend & Embedded Software Engineer with expertise in **C#, .NET, C++ and SQL**. Design and develop reliable backend systems and embedded software for Linux-based IoT devices with a focus on clean architecture, performance and scalability.

Skills

Languages & Frameworks: C#, C++, .Net, ASP.NET, Node.js, SQL, MQTT, CMake.

Backend & Systems: REST APIs, Windows Services, Database Design, System Design.

Cloud: AWS Services, Lambdas, S3, IoT Core, SQS.

DevOps & Tools: Git, CI/CD, Docker, Kubernetes, Grafana, Jira, Agile.

Frontend/UI (familiar): WinForms, WPF, Razor Pages, Angular, React, Flutter.

AI-assisted Software Development: Daily use of modern LLM tools (**GitHub Copilot, Cursor and OpenAI API**) to accelerate development, refactoring, debugging and technical research. Practical knowledge of **LLM integrations, RAG concepts, embeddings** and **vector databases**.

Experience

Backend & Embedded Software Engineer | Galcon Computerized Control Systems April 2023 – Present

Develop backend services and embedded software for smart farm IoT controllers using **C#, C++, .Net, Node.js, SQL Server, MQTT, Redis and AWS Cloud Services**

- Building and optimizing **REST APIs** and backend communication systems
- Designing and maintaining database structures and system-level architecture in collaboration with product and technical leadership
- Played a leading role in a major embedded architecture refactoring replacing native TCP socket with **MQTT broker** and migrating configuration management from device registers to a database-driven architecture
- Designing and developing embedded software for Linux-based IoT devices using OOP principles and clean architecture, with a strong focus on performance, concurrency and thread-safety
- Leveraging AI-assisted development tools to improve development efficiency, technical research and architectural decision-making
- Acting as a technical point of contact within an international development team, helping engineers clarify requirements, discuss implementation approaches and make technical decisions
- Working in an Agile team using Git and Jira
- Built a scalable load-testing tool to simulate thousands of IoT devices
- Developed a desktop tool using WinForms to track and simulate Redis messages

Full Stack Developer | Minicy Catom LTD

August 2021 – March 2023

Developed and maintained an energy control system using **C#, ASP.Net and SQL Server**

- Designed database structure
- Optimized SQL procedures, reducing report and billing generation time
- Developed Windows Services for background tasks

Education

Full Stack Developer | John Bryce, Haifa

March 2020 – April 2021

Graduated with honours

Programming Engineer | ORT Braude, Karmiel

October 2018 – March 2020

Paused due to personal circumstances

Electro-mechanical Junior Engineer | ORT Braude, Karmiel

October 2015 – June 2017

Graduated with honours

Projects

WorkingHoursAPI | 2024 – 2025 | [Git](#)

Technologies: C#, PostgreSQL

Built a REST API for tracking work shifts. Practiced PostgreSQL management with pgAdmin and data access using Entity Framework Core.

CMake Project | 2022 | [Git](#)

Technologies: CI/CD Actions, CMake, Docker, C++

Created a cross-platform C++ project using CMake and Docker with automated CI/CD pipelines.

(More projects on [GitHub](#))

Languages

Hebrew – Native

Russian – Native

English – Proficient